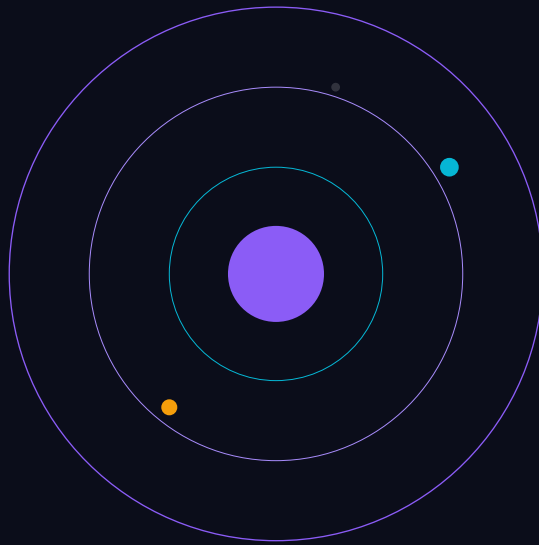




Zenith Quest

Your Personal Sky Guide

Technical Proposal — Hackathon Round 1

PREPARED BY

Zenith Quest Team

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7 Live APIs

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3D Sky Dome

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Gamified Missions

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AI Narrator

June 2026

1. Problem Statement

Millions of people look up at the night sky and wonder — what's that bright star? When does the ISS fly over? Is tonight good for stargazing? Yet existing astronomy apps are either too technical for casual users, or too shallow to be useful. There is no single platform that answers the simple question: "What's happening in MY sky tonight?"

The Core Challenges

- No unified platform — Users juggle separate apps for weather, satellite tracking, planet charts, and moon phases.
- Overwhelming complexity — Planetarium apps show thousands of objects with no personalized guidance or priority.
- No "sky quality" metric — Users have no way to quickly know if tonight is worth going outside for stargazing.
- Zero engagement loop — Passive star charts offer no motivation for repeated use or deeper exploration.
- Data fragmentation — Weather, celestial positions, satellite passes, and solar activity exist across 7+ disconnected APIs.

2. Our Solution — Zenith Quest

Zenith Quest is a real-time celestial exploration platform that transforms the night sky into a personalized, interactive experience. It combines live data from 7+ astronomical APIs, an AI-powered narrative engine, immersive 3D visualizations, and a gamified mission system into one seamless sky-watching companion.

What Makes It Unique

- Sky Score Engine — A proprietary 0–100 composite rating using 6 weighted factors (cloud cover, light pollution, planets, satellites, moon, transparency).
- AI Sky Narrator — GPT-4o-mini synthesizes all live data into natural-language summaries: "Tonight is excellent for stargazing! Venus and Jupiter are visible..."
- Interactive Satellite Tracker — Real-time Leaflet map with ISS, Starlink, and Hubble positions and orbit paths.
- 3D Sky Dome — An immersive Three.js celestial sphere showing planets, constellations, and the moon in your local sky.
- Celestial Timeline — A chronological guide of every sky event from sunset to sunrise.
- Gamified Missions & XP — Observation, tracking, and identification missions with XP, 7 explorer ranks, badges, and daily challenges.
- NASA Integration — Astronomy Picture of the Day, Near Earth Objects, and Solar Flare activity.

3. Technology Stack

% Frontend Framework

- Next.js 16 (App Router, SSR, Turbopack)
- React 19 (Server Components, Suspense)
- TypeScript 5 (20+ type interfaces)

% Visualization

- Three.js + React Three Fiber — 3D sky dome
- Leaflet + React-Leaflet — satellite tracking map
- Recharts — analytics dashboards

% Styling & Animation

- Tailwind CSS v4 — utility-first dark theme
- Framer Motion — page transitions & micro-animations

% State & Data

- Zustand — client state (location, sky, user, missions)
- SWR — data fetching with caching & revalidation

% Backend & Auth

- Clerk — OAuth authentication (Google, GitHub)
- MongoDB + Mongoose — user data & observation history
- Next.js API Routes — 10 server endpoints

4. APIs & Data Sources

Weather

Open-Meteo

Cloud cover, visibility, temperature, wind, humidity, hourly forecast, sunrise/sunset. Free, no auth. Refresh: 15 min.

Satellites

N2YO

Real-time ISS/Starlink/Hubble positions, visual pass predictions, satellites-above-observer. API Key. Refresh: 1 sec.

Celestial

AstronomyAPI

Planet positions (altitude, azimuth, magnitude, constellation), moon phase & illumination. Basic Auth. Refresh: 1 hr.

Space Data

NASA APIs

Astronomy Picture of the Day (APOD), Near Earth Objects (NeoWs), Solar Flares (DONKI). API Key. Refresh: 1 hr.

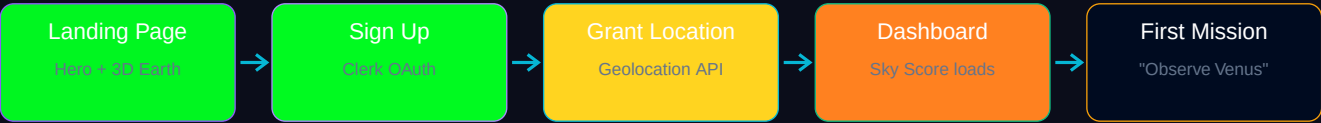
AI

OpenAI

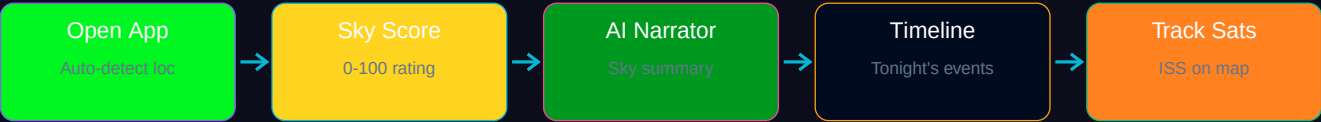
GPT-4o-mini generates natural-language sky narratives. Falls back to deterministic template engine. Bearer token.

6. User Flow Diagrams

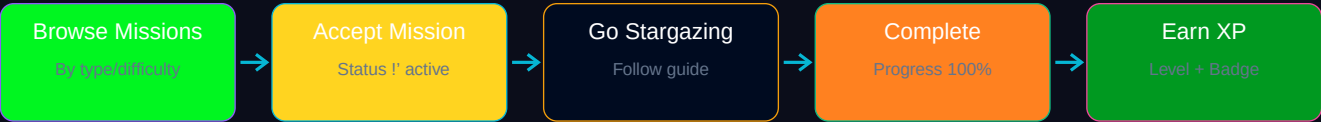
Flow 1: First-Time User Onboarding



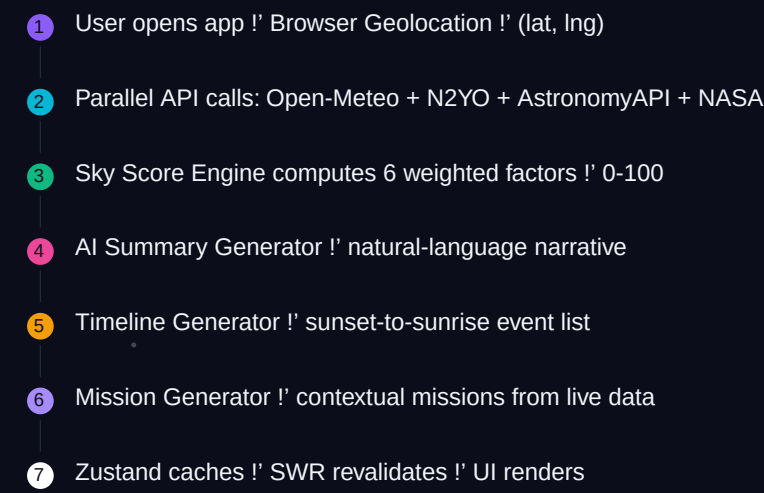
Flow 2: Core Sky Exploration Loop



Flow 3: Gamification Loop

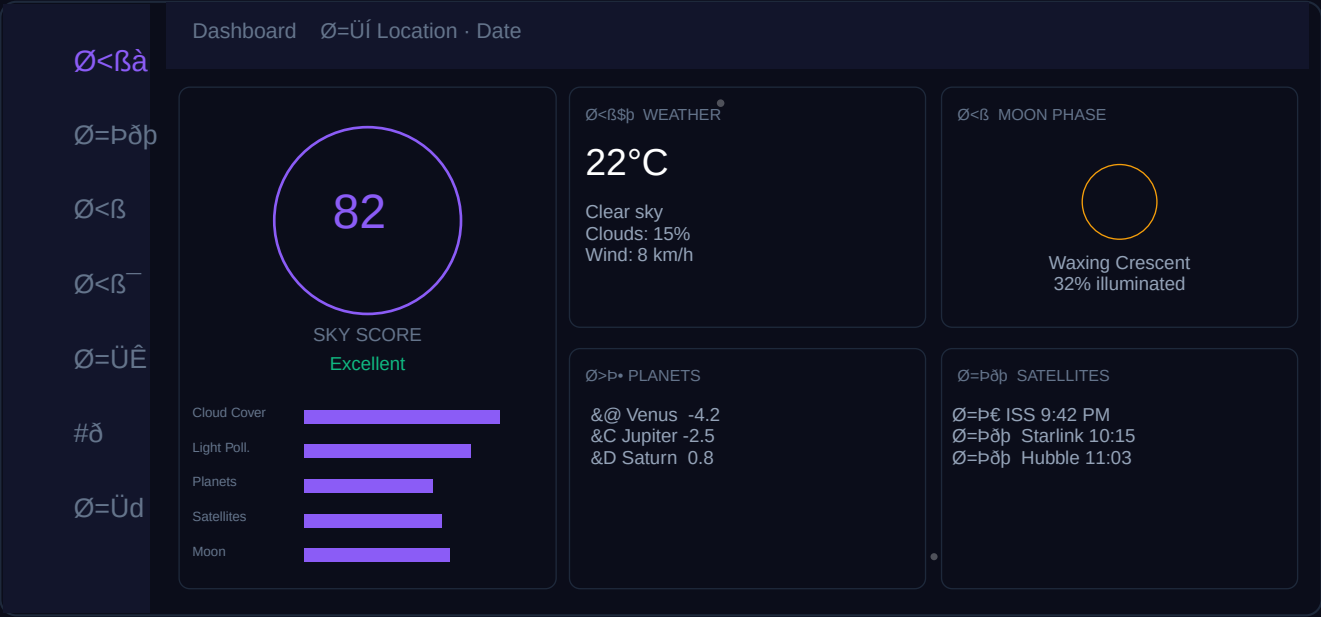


Flow 4: Real-Time Data Pipeline

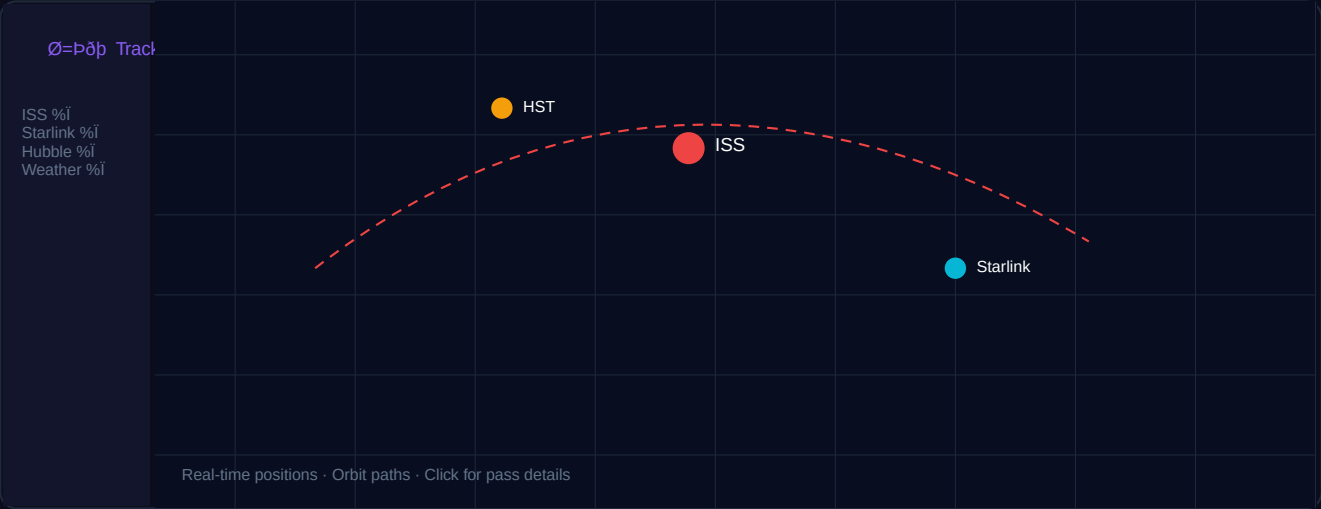


7. Low-Fidelity Wireframes

Dashboard — Main View



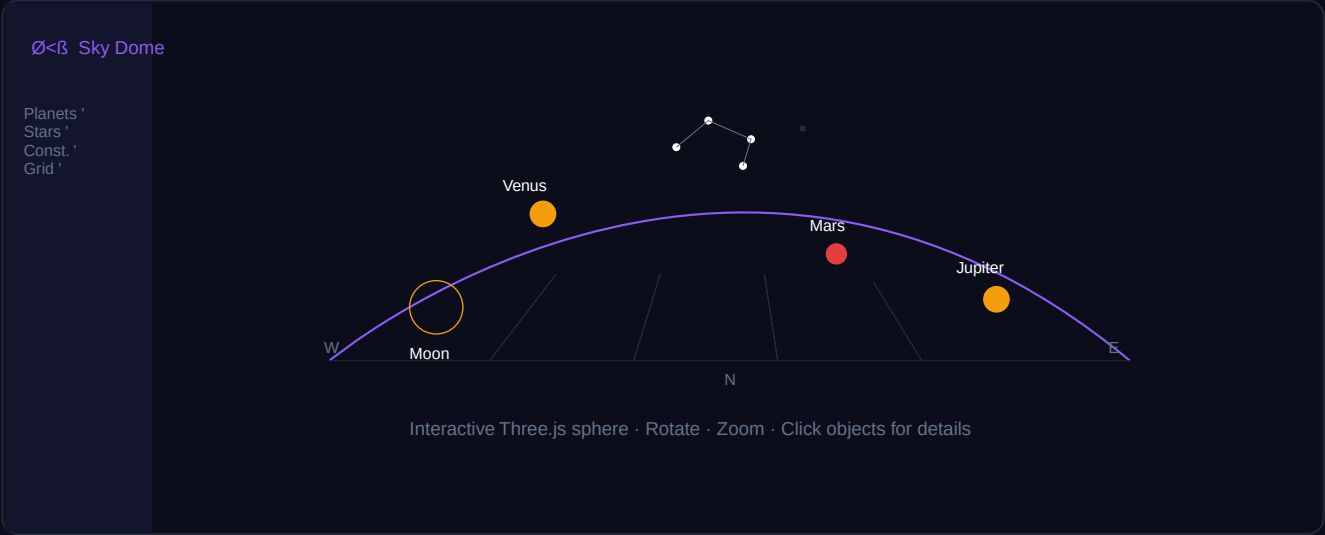
Satellite Tracker — Map View



8. Sky Score Engine — Algorithm Diagram



3D Sky Dome — Wireframe



9. Implementation Strategy

Days 1–2

Phase 1 — Foundation

- Next.js 16 project setup with TypeScript, Tailwind CSS v4, Turbopack
- Clerk authentication + MongoDB Atlas database schemas
- Dark-mode design system: glassmorphism cards, gradients, cosmic palette
- Define 20+ TypeScript interfaces for all data models

Days 3–4

Phase 2 — Data Engine

- Integrate all 5 APIs: Open-Meteo, N2YO, AstronomyAPI, NASA, OpenAI
- Build Sky Score Engine with 6-factor weighted algorithm
- Create comprehensive mock data for full demo-mode operation
- Build AI Summary Generator with deterministic fallback

Days 5–7

Phase 3 — Visualizations

- Dashboard: SkyScoreGauge, WeatherCard, MoonPhaseCard, PlanetCard, SatelliteCard
- Satellite Tracker: Leaflet map with ISS/Starlink markers + orbit paths
- 3D Sky Dome: Three.js hemisphere with planets and constellations
- Celestial Timeline + Landing page with 3D Earth hero

Days 8–9

Phase 4 — Gamification

- Mission Generator: contextual observation/tracking/identification missions
- XP system with 7 explorer ranks (Stargazer !' Universe Sage)
- Badge system: 5 rarity tiers × 6 categories + daily challenges

Days 10–12

Phase 5 — Polish & Deploy

- Analytics dashboard with Recharts, responsive testing
- Performance: ISR caching, dynamic imports, bundle optimization
- SEO, accessibility review, Vercel deployment, demo recording

10. Conclusion

Zenith Quest addresses a clear gap in the astronomy app market by combining real-time data from 7 APIs, an AI-powered narrative engine, immersive 3D visualizations, and gamification into a single, beautiful platform. The modular architecture ensures scalability, the mock data system guarantees a flawless demo experience, and the dark-first design creates an immersive stargazing companion.

With a clear implementation timeline, proven technology stack, and strong feasibility profile, Zenith Quest is ready to transform how people experience the night sky.
